



Battery Management & Charging

Li-ion safety, BMS, gauges, USB-C PD

Chemistry • Charging • Safety • Gauging • Power-delivery

Audience: Intermediate Engineering Students

Cells • Stacks • Protection • Charging ICs • USB-C PD



Why Battery Engineering Matters

Get it wrong — get a fire

Safety critical

Li-ion failure modes include thermal runaway.

Compliance

UN 38.3, IEC 62133, BIS required for shipping.

Battery life

Charging strategy determines cycle life by 5-10x.

Run-time

Capacity utilization + low-power MCU sleep.

Cost

Battery + protection + fuel-gauge $\approx$ 20% of BOM.

Returns / RMA

Defective batteries = biggest support cost.

Battery Chemistries

Different chemistries, different rules

Chemistry	Nominal V	Energy density	Use
Alkaline (AA)	1.5 V	100 Wh/kg	Remote controls, low-drain
NiMH	1.2 V	120 Wh/kg	Toys, cordless drills
Li-ion (cylindrical)	3.6/3.7 V	250 Wh/kg	Laptops, e-bikes, EVs
Li-Po (polymer)	3.7/3.8 V	260 Wh/kg	Phones, drones, wearables
LiFePO ₄ (LFP)	3.2 V	150 Wh/kg	Safer, longer life, EVs
NMC / NCA	3.6-3.7 V	280 Wh/kg	EVs, premium products
Solid-state	3.6 V	350+ Wh/kg	Future, low commercial use
Lead acid (SLA)	2 V/cell	35 Wh/kg	UPS, automotive starter

Li-Ion Safety — Why It's Different

The narrow operating window

- ▶ Overcharge → > 4.25 V/cell → exothermic decomposition → fire
- ▶ Over-discharge → < 2.5 V/cell → permanent capacity loss
- ▶ Over-current → internal heating → separator melts → short → fire
- ▶ Over-temperature → > 60 °C charging, > 70 °C discharging → fire
- ▶ Mechanical damage → puncture/dent → internal short → fire
- ▶ Fire is the same word every time — Li-ion is energy-dense; treat it like fuel
- ▶ EVERY commercial Li-ion product needs a protection circuit (PCB / PCM / BMS)

Li-Ion Charging Profile

CC-CV (Constant Current then Constant Voltage)

- ▶ Pre-charge: < 3.0 V/cell — small current (0.1C)
- ▶ Constant Current (CC): 0.5-1C charge rate, voltage rising
- ▶ Constant Voltage (CV): voltage clamped at 4.2 V, current tapering
- ▶ Termination: current drops to 0.05-0.1C → battery full
- ▶ Standby: monitor voltage; trickle if cell drops below 4.05 V
- ▶ C-rate definition: C = capacity in Ah; 1C = 1 hour discharge

Temperature dependency

Don't charge below 0 °C — plating damages anode permanently. Don't charge above 45 °C — accelerates aging.

Cycle life

Charging to 100% reduces life vs 80%. EVs limit upper cutoff to ~95% for longevity. Storage at 50% SOC, 15 °C is ideal.

Charging IC Selection

Pick the right charger for the cell count + current

IC	Input	Cells	Charge I	Notes
MCP73831	USB 5 V	1 cell	≤ 500 mA	Single Li-ion, no I ² C
TP4056	USB 5 V	1 cell	≤ 1 A	Ubiquitous, cheap
BQ24074	5-7 V	1 cell	≤ 1.5 A	USB-friendly, power-path
BQ25890	3.9-14 V (USB-PD)	1 cell	≤ 5 A	Fast charging, I ² C
BQ25700A	4-24 V	1-4 cells	≤ 8 A	Buck-boost charger
LTC4079	4-60 V	Any	≤ 250 mA	Industrial low-power

Power-path

Power-path chargers let the system run from USB while battery charges. Without power-path, plugging in disconnects load briefly.

Protection Circuit (PCM / PCB)

What sits between the cell and the world

- ▶ Two back-to-back N-MOSFETs in series with cell negative
- ▶ Comparator IC monitors cell voltage and current
- ▶ Triggers: overvoltage, undervoltage, over-current charge/discharge, short circuit
- ▶ Common ICs: DW01 (cheapest, 1 cell), S-8261, BQ29700 (TI)
- ▶ Multi-cell: AFE (Analog Front End) — BQ76920, BQ76952, ISL94203
- ▶ Most 18650 cells ship with protection PCM built into the cell sleeve
- ▶ Cell balancing: passive (resistor) or active (capacitor / inductor)

Fuel Gauges (State-of-Charge)

How much battery is left?

Voltage only

Simple ADC + lookup table.
Drift at end-of-life.

Coulomb counter

Integrate current in/out.
Accurate, needs calibration.

Impedance track

Adapts to cell aging.
TI patented technique.

Common ICs

MAX17048 (V-only), MAX17260,
BQ27220 (Impedance Track).

Calibration

Full charge & discharge once
teaches the gauge.

Self-discharge

Cells lose 2-5% per month.
Account for in firmware.

USB-C Power Delivery (PD)

Up to 240 W over a USB-C cable

- ▶ USB-C connector + PD protocol → negotiated voltage 5-48 V
- ▶ PD 3.0: up to 100 W ($20\text{ V} \times 5\text{ A}$); PD 3.1 EPR: up to 240 W ($48\text{ V} \times 5\text{ A}$)
- ▶ PPS (Programmable Power Supply): step in 20 mV / 50 mA for charging
- ▶ CC1 / CC2 pins: orientation + role detection + PD signalling
- ▶ Sink advertises capabilities; source picks; negotiation < 1 s
- ▶ Cheap PD sink ICs: TPS25750, FUSB302, IP2721, CH224K
- ▶ EMI: USB-C with PD must pass CISPR 22 — careful CMC + bypass design

Multi-Cell Packs

When one cell isn't enough

- ▶ Series for voltage: 2S = 7.4 V, 3S = 11.1 V, 4S = 14.8 V
- ▶ Parallel for capacity: 2P doubles Ah, halves internal R
- ▶ EV packs: 96S + 4P = 350 V at 80 Ah, ~28 kWh
- ▶ Cell balancing essential — even 5 mV imbalance compounds over cycles
- ▶ BMS measures every cell separately — no shortcuts
- ▶ Cell selection: closely matched IR + capacity from same batch
- ▶ Thermal sensors per group of cells
- ▶ Pre-charge resistor to limit inrush when connecting big caps

Battery-Powered Power Tree

From cell to clean rails

- ▶ 1 cell Li-ion (3.0-4.2 V) → buck-boost → 3.3 V system rail
- ▶ 3.3 V → LDO → 1.8 V analog and sensor rail
- ▶ Boost → 5 V for USB host / motor
- ▶ Charger handles USB input AND charging in same IC
- ▶ Power-path: load runs from USB OR battery automatically
- ▶ Sleep modes: drop unused rails to save μA
- ▶ Low-IQ regulators: TPS62840 (60 nA), MP2155 (low-IQ buck)

Common Battery-Design Mistakes

What field returns reveal

- ▶ No protection circuit on bare cell — fire hazard
- ▶ Charging without temperature sensor — overheats in summer
- ▶ Single MCU pin reading cell V — no isolation, noisy reads
- ▶ Charging IC without thermal pad copper area — shuts down
- ▶ Multi-cell without balancing — pack capacity = weakest cell
- ▶ USB-PD sink without CC pull-down — never enumerates
- ▶ Big bulk cap on USB-PD input without inrush limit
- ▶ No deep-sleep — battery dies in months on the shelf

Battery Compliance

Certifications required to ship

Standard	What it covers	Cost (₹)	Region
UN 38.3	Battery shipping (transport)	1-3 L	Global
IEC 62133	Portable cell safety	2-5 L	Global
BIS IS 16046	Cell safety, India	1-3 L	India
UL 2054	Battery pack safety (USA)	3-8 L	USA
UL 1642	Single cell, USA	2-4 L	USA
UN 38.3 + MSDS	Air shipping	₹50k+	Global

India context

BIS certification is mandatory for Li-ion batteries sold in India since 2021. Without it, customs can seize shipments.

Best Practices

Habits of a careful battery designer

- ▶ Always specify a protection circuit
 - ▶ Add NTC thermistor near every cell group
 - ▶ Use a charger IC with JEITA temperature curve
 - ▶ Coulomb counter for accurate gauge
 - ▶ Pre-charge mode for deeply-discharged cells
 - ▶ Soft-start on power-up to limit inrush
- ▶ Plan for ageing: 80% capacity at end-of-life
 - ▶ Periodic full-cycle re-calibration in firmware
 - ▶ Log charge / temperature history
 - ▶ Use cells from reputable vendors (LG, Samsung, Panasonic)
 - ▶ Cycle test before shipping to customers
 - ▶ Document worst-case fault modes for review

Key Takeaways

Key takeaways from this lecture

Safety first

Li-ion failure = fire.

Protection mandatory

PCM in every product.

CC-CV charging

Pre-charge → CC → CV → trickle.

Temperature window

0-45 °C charge, < 60 °C discharge.

Multi-cell = balancing

Cell-by-cell measurement.

Compliance early

UN 38.3, IEC 62133, BIS.

Recommended Resources

Battery learning paths

- ▶ TI Battery Management University (free online)
- ▶ BatteryUniversity.com — chemistry & care info
- ▶ Davide Andrea — 'Battery Management Systems for Large Li-Ion Battery Packs'
- ▶ Linden / Reddy — 'Handbook of Batteries'
- ▶ MAX / TI / NXP application notes on BMS design
- ▶ USB-IF Power Delivery Specification (free)
- ▶ BIS / IEC compliance guides
- ▶ Hackaday / Electroboom YouTube — practical battery stories

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- ▶ TI Battery Management University
- ▶ BatteryUniversity.com — chemistry and care
- ▶ Davide Andrea — Battery Management Systems
- ▶ USB-IF Power Delivery spec
- ▶ BIS / IEC certification guides
- ▶ MAX / TI / NXP BMS application notes
- ▶ Hackaday battery teardown threads
- ▶ Local certification labs: SGS, Intertek (India)

Thank you • Build something this week.